

# Quantum Gravitation

## Lecture 3

# Canonical Quantization

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

## ■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$[q_{ks}, p_{k's'}] = i\hbar\delta_{kk'}\delta_{ss'}$$

**■ Quantization**

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$H = \sum_{k,s} \hbar \omega_{ks} (a_{ks}^\dagger a_{ks} + 1/2)$$

**■ Spectrum and response**

Poles, residues, and group velocity are read from the same quadratic operator.

$$D_{ij,kl}(x) = \langle 0 | T h_{ij}(x) h_{kl}(0) | 0 \rangle$$

## ■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

**■ Example**

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 3; Cole, ch. 4; Q-Medium notebook 03